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Modifications to **sicegar**: Analysis of Single-Cell Viral Growth Curves

Samuel Butler[1*], **Phineus Choi**[1*], **Thomas Matheis**[1*], **Mira Terdiman**[1], and **Johanna Hardin**[1]

1 Pomona College, United States * These authors contributed equally.

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Summary

The R package **sicegar** aims to quantify time-intensity data by using sigmoidal and double-sigmoidal curves. The method was originally developed in the context of single-cell viral growth analysis (for details, see Caglar et al. (2018)), and the package name stands for "SIngle CEll Growth Analysis in R". In addition to categorizing the best model fit, **sicegar** returns parameter estimates for each of the model fits, providing important information to researchers.

The sigmoidal function is given as $I(x)$, with parameters estimated using the Levenberg-Marquardt algorithm (Levenberg, 1944; Marquardt, 1963). h_0 represents the lower asymptote (as x approaches negative infinity). t_1 is the onset time (the inflection point) and the midpoint between h_0 and h_1. a determines the magnitude of the slope of the sigmoidal curve. h_1 is the upper asymptote (as x approaches positive infinity). The parameters describing the sigmoidal curve in $I(x)$ are graphically represented in **Figure 1**.

$$I(x) = h_0 + \frac{h_1 - h_0}{1 + e^{-a(x - t_1)}}$$

A similar, but slightly more complicated, formula for the double-sigmoidal function is also used for parameter estimation in **sicegar**. In the original implementation of **sicegar**, the parameter h_0 is set to zero for both the sigmoidal and double-sigmoidal models.

Figure 1 Visualization of the sigmoidal curve and related parameter values.

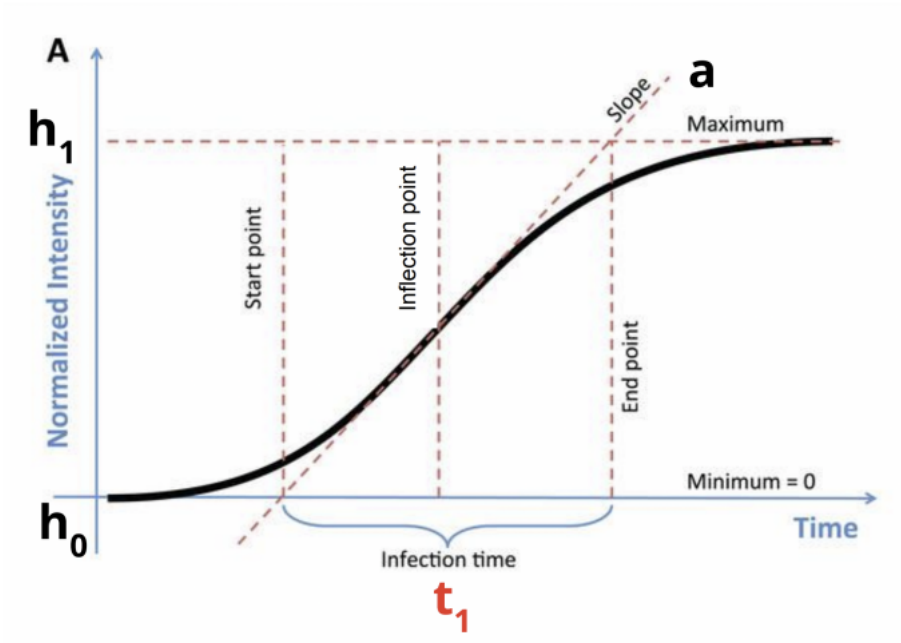

Figure 1: Sigmoidal curve.

sicegar's parameter estimation allows users to extract key model parameters, including the onset time of RNA expression in specific genes (Caglar et al., 2018). Onset time is defined as the time when the half-maximal abundance of RNA is reached in the cell. For example, Adams et al. (2023) uses **sicegar** to investigate the onset time of RNA expression from genes in *E. coli* undergoing stress. Accurate estimates of t_1 are crucial for understanding transriptional timing of genes. However, Adams et al. (2023) observed limitations in **sicegar**'s fits, which motivated the package improvements proposed in our paper.

We have substantially modified **sicegar** (our work is currently implemented on CRAN) to improve the parameter estimates for both sigmoidal and double-sigmoidal models. Our primary update to the **sicegar** package is the inclusion and estimation of an additional parameter, h_0, the lower asymptote of both the sigmoidal and double-sigmoidal curves, which had previously been set equal to zero. Based on simulated data, the free estimation of h_0 provides both a better overall model fit (lower SSE) and more accurate parameter estimates than when h_0 is forced to be equal to zero.

Statement of Need

Caglar et al. (2018) discuss **sicegar**'s ability to correctly identify sigmoidal and double-sigmoidal curves on simulated data. They report, "Overall, we can conclude that our algorithm results in reliable fits, that it fails gradually with increasing noise levels, and that it is conservative in assessing whether it has correctly identified a sigmoidal or double-sigmoidal curve or not." The focus of their results is on whether **sicegar** is able to correctly identify sigmoidal and double-sigmoidal curves on simulated data, not on whether it is able to accurately estimate parameters. Although they acknowledge **sicegar**'s ability to estimate parameters associated with sigmoidal and double-sigmoidal curves, they do not report on the accuracy of the parameter estimates.

In dozens of papers that use **sicegar** for modeling time-intensity data, researchers are interested in extracting specific parameter estimates, e.g., midpoints and slopes, because the parameter values represent biologically meaningful information. Adams et al. (2023) extracted t_1 to

investigate onset time of RNA expression in genes in *E. coli* undergoing stress. Wittemeier et al. (2025) used **sicegar** to estimate molar carbon assimilation, using **sicegar**'s estimate of a to understand maximum assimilation rate and t_1 to extract the point at which maximum assimilation is reached. Rajarathinam et al. (2025) also used **sicegar** to analyze carbon assimilation and to extract estimates of a. Our addition of h_0 to the set of estimated parameters greatly improves **sicegar**'s ability to provide accurate parameter estimates (note, if one parameter is wrong it impacts the entire fit of the model). Though the categorization of the model as sigmoidal or double-sigmoidal is hugely important, it is not the only important aspect of the **sicegar** modeling. Our updated implementation, which includes the estimate of the lower asymptote, is prevailingly important, and thus our adjustments fit the needs of current research.

Features

Core Functions

`fitAndCategorize`, the umbrella function in **sicegar**, takes time-intensity data as an argument and runs the data through a series of nested functions. The process is outlined in **Figure 2**. First, the data are normalized in the function `normalizeData` (1). Then they are passed through `multipleFitFunction` (2) which uses the Levenberg-Marquardt algorithm (Levenberg, 1944; Marquardt, 1963) to fit both sigmoidal and double-sigmoidal curves to the data, (3), (4). Additional parameters are estimated using `parameterCalculations`, and the model is visualized using `figureModelCurves`. The user decides whether to allow the function to estimate h_0 using the argument `use_h0 = TRUE` in `fitAndCategorize`. If `use_h0` is set to `FALSE` (the default, for backward compatibility), the algorithm will run **sicegar** using the original implementation, with h_0 fixed at zero. If the user chooses to estimate h_0, the same function flow is followed, and each sub-function will account for the estimate of h_0.

Novel Contributions

Each function in **sicegar** was rewritten to include the parameter h_0. Our new version is outlined in the right-hand branch of **Figure 2**.

Figure 2 Structure of the `fitAndCategorize` function.

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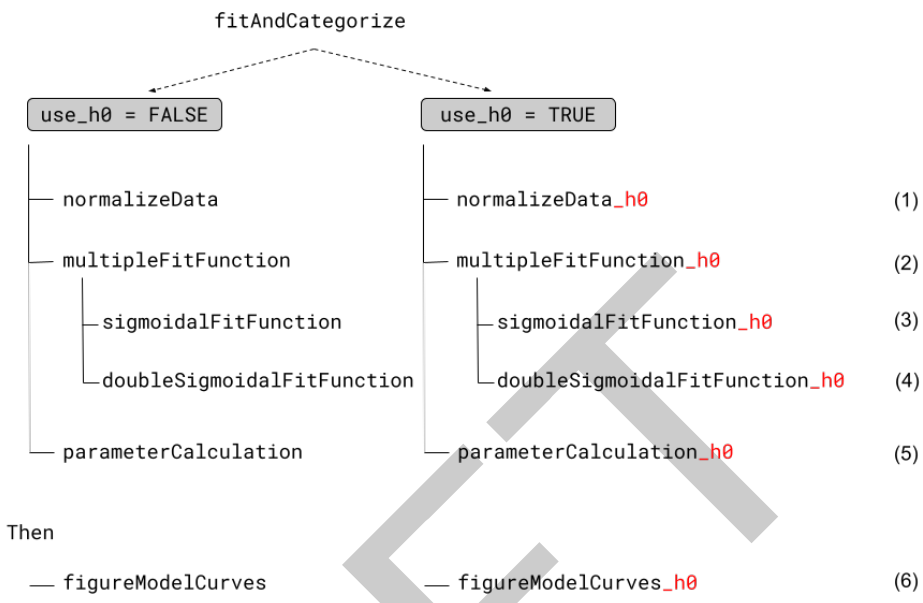

Figure 2: Structure of the `fitAndCategorize` function.

Example

To demonstrate the difference between the functionality when $h_0 = 0$ versus when h_0 is freely estimated, we present a small simulation based on sigmoidal data. The parameters in our simulation are $h_0 = 2$, $h_1 = 4$, $a = 1$, and $t_1 = 12$. At each of nine time points on the x-axis, we have five of replicates around the sigmoidal model with normal noise (centered at 0, standard deviation of 3).

Figure 3 Parameter estimates from simulated data.

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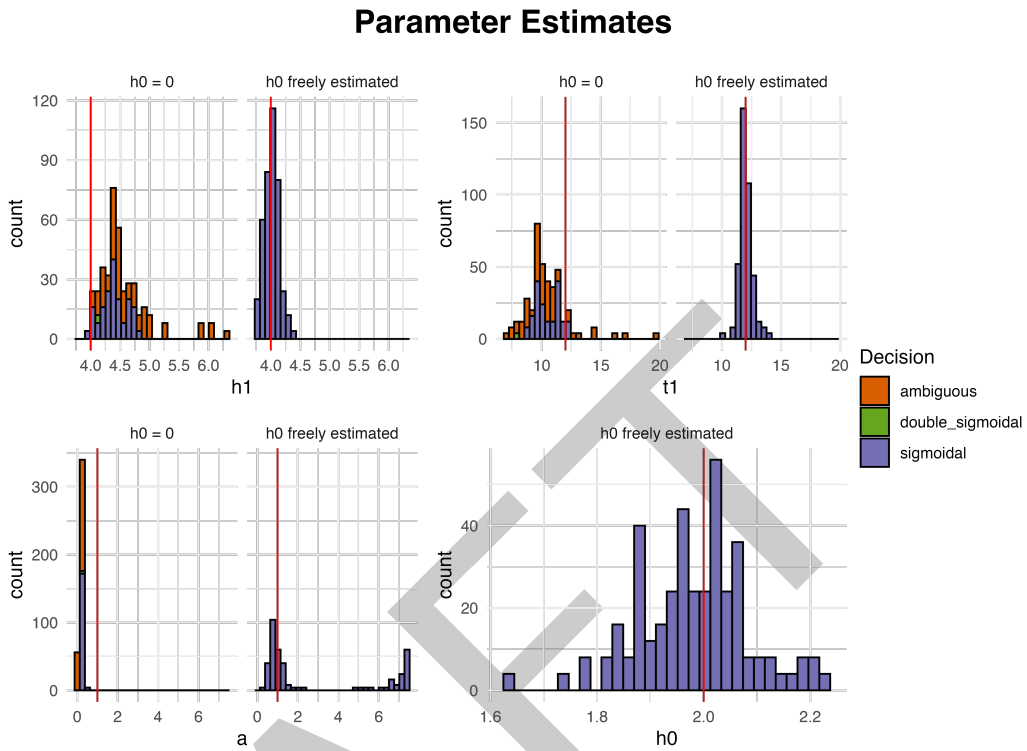

Figure 3: Parameter estimates from simulated data.

For each of 200 simulated sigmoidal datasets, **Figure 3** shows the parameter estimates for each of the four parameters given in $I(x)$, see **Figure 1**. The red vertical line indicates the value of the parameter used for data generation. It should be noted that allowing h_0 to be freely estimated provides both more accurate parameter estimations — notably for t_1 — and better categorizations of the model (as "sigmoidal" rather than "ambiguous"). Similar results occur when the package is run on simulated data with more noise, as well as on data simulated from the double-sigmoidal model. Our simulation does not prove that estimating h_0 freely is always preferable, but it does indicate that there are circumstances in which estimating h_0 is important for full approximation of the model.

Availability

The **sicegar** package is available on CRAN (https://CRAN.R-project.org/package=sicegar) and GitHub (https://github.com/hardin47/sicegar). Documentation, including vignettes and examples, is provided to facilitate adoption.

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